

DIGITAL FORENSICS

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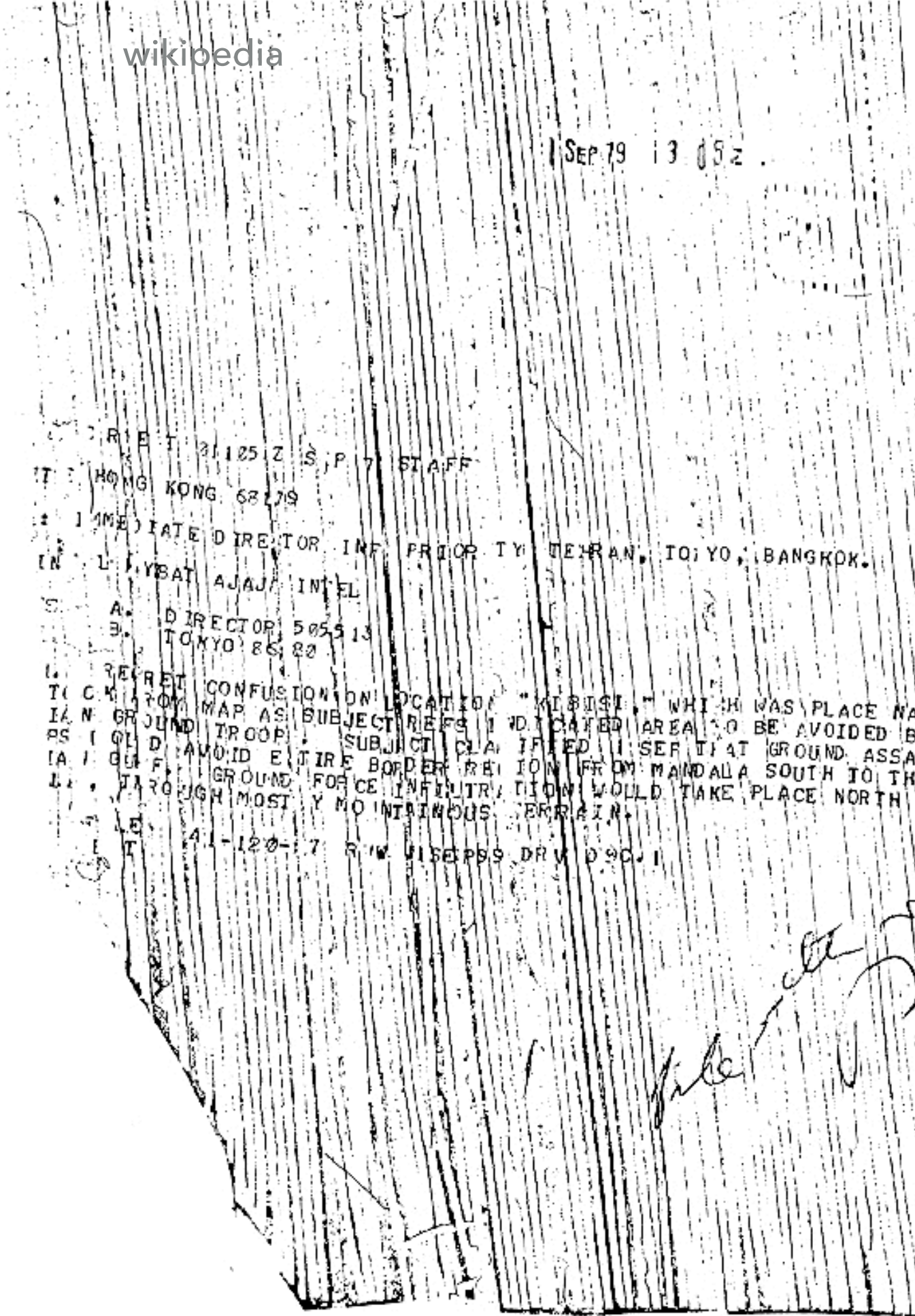
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PHYSICAL EVIDENCE

HOW EVIDENCE GETS DESTROYED



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WHAT IS DIGITAL FORENSICS

- ▶ We define computer forensics as the discipline that combines elements of law and computer science to collect and analyze data from computer systems, networks, wireless communications, and storage devices in a way that is admissible as evidence in a court of law. [US-Cert]
- ▶ Consists of
 - ▶ Networks (Network Forensics)
 - ▶ Small Scale Digital Devices
 - ▶ Storage Media (Computer forensics)
 - ▶ Code Analysis

CAN DIGITAL EVIDENCE BE DESTROYED?

Digital Detectives (2011) - Documentary on Computer Forensics in the DoD

<https://www.youtube.com/watch?v=QKebBYsaMgU>

FORENSICS PROCESS

- Acquisition or imaging of exhibits
 - Using dd, dc3dd, FTK Imager, etc.
 - Created, hashed, and write-blocked to prevent changes
- Analysis
 - Visible content such as folders on the desktop and images in user files
 - Hidden, encrypted, or deleted files
- Reporting

IMAGE TAMPERING

NIST Colloquium Series: Digital Forensics

by Dr. Hany Farid

<https://www.youtube.com/watch?v=9DKJ6gP5IJY>



SPECULARITY

NIST



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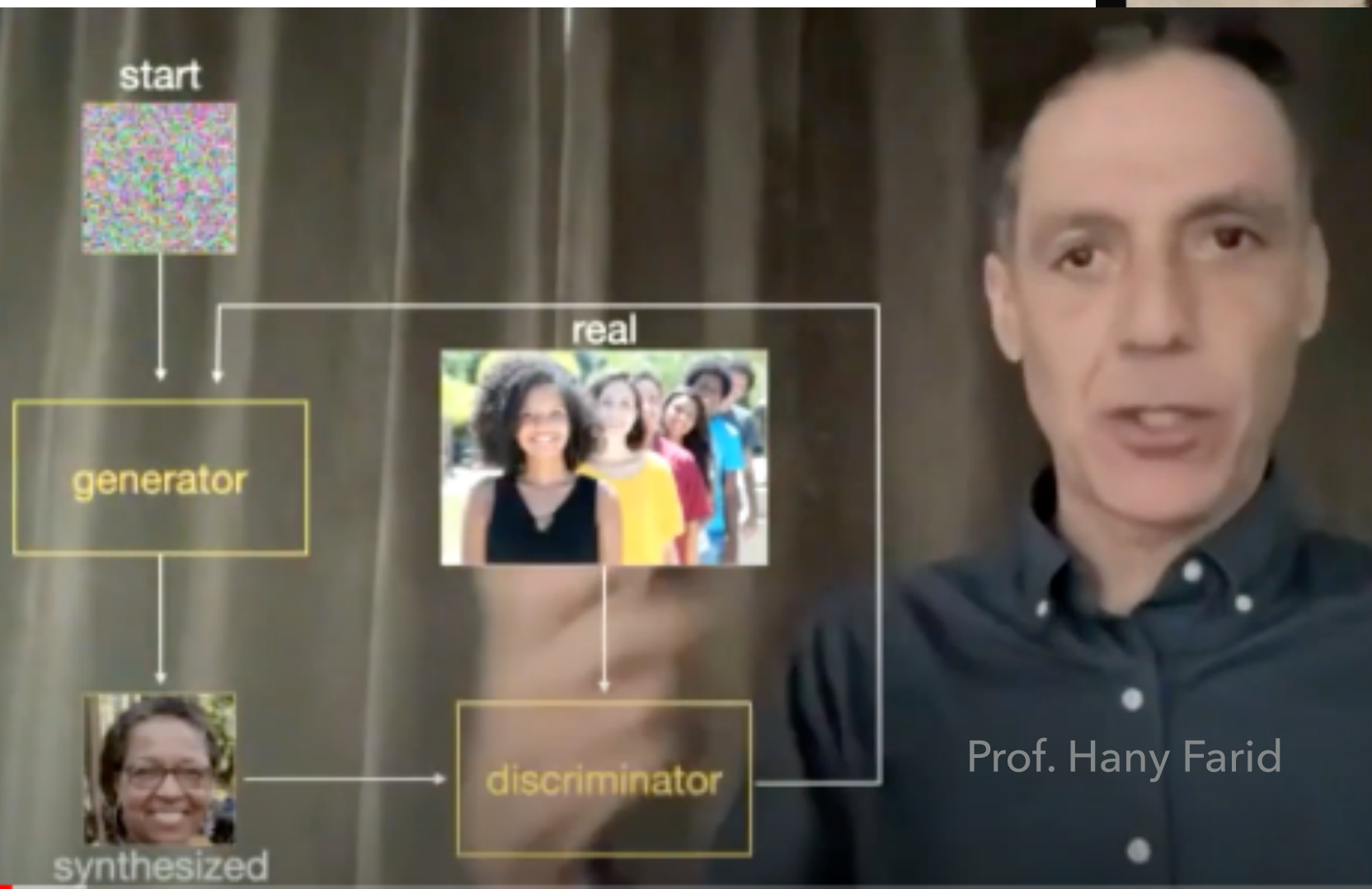
HMMM?



This is a **retouched picture**, which means that it has been digitally altered from its original version. Modifications made by [Mmxx.kunwadee \(AT\) cp.eng.chula.ac.th](#)

DEEP FAKES AND GANS

https://www.youtube.com/watch?v=s_mWRJJHdKI





HMMM...





MELISSA VIRUS (1999)

Path: [newsflash!news-peer1.tiac.net!news-feed1.tiac.net!newshub.northeast.verio.net!cpk-news-hub1.bbnplanet.com!news.gtei.net!newsfeed.cwix.com!152.163.199.19!portc03.blue.aol.com!audrey01.news.aol.com!not-for-mail](#)
From: [skyr...@aol.com](#) (Sky Rocket)
Newsgroups: alt.sex
Subject: Passcode List 3-26-99
Lines: 283
NNTP-Posting-Host: [ladder05.news.aol.com](#)
X-Admin: [ne...@aol.com](#)
Date: 26 Mar 1999 12:15:53 GMT
Organization: AOL <http://www.aol.com>
Message-ID: <[19990326071553.24526.00000525@ng-cg1.aol.com](#)>
Xref: newsflash alt.sex:1680745

_=
= Part 001 of 001 of file list.zip
=

This message contains a zip file with the now famous list.doc file in it. The file is infected with the Melissa virus.

MELISSA VIRUS (1999)

I did some "dumpster diving" in the list.doc file and found the following revision log in the file:

```
3360:07 00 FF FF 06 00 00 00 0B 00 4A 00 6F 00 68 00 .....|..J.o.h.
3370:6E 00 20 00 48 00 6F 00 6C 00 6D 00 65 00 73 00 n. .H.o.|l.m.e.s.
3380:19 00 43 00 3A 00 5C 00 57 00 49 00 4E 00 44 00 ..C.:|. |W.I.N.D.
3390:4F 00 57 00 53 00 5C 00 44 00 65 00 73 00 6B 00 O.W.S.|. |D.e.s.k.
33A0:74 00 6F 00 70 00 5C 00 50 00 30 00 2E 00 64 00 t.o.p.|. |P.O...d.
33B0:6F 00 63 00 0B 00 4A 00 6F 00 68 00 6E 00 20 00 o.c...J.|o.h.n. .
33C0:48 00 6F 00 6C 00 6D 00 65 00 73 00 1F 00 43 00 H.o.l.m.|e.s...C.
33D0:3A 00 5C 00 57 00 49 00 4E 00 44 00 4F 00 57 00 :|. |W.I.|N.D.O.W.
33E0:53 00 5C 00 44 00 65 00 73 00 6B 00 74 00 6F 00 S|. |D.e.|s.k.t.o.
33F0:70 00 5C 00 4C 00 69 00 73 00 74 00 30 00 38 00 p|. |L.i.|s.t.O.8.
3400:31 00 39 00 2E 00 64 00 6F 00 63 00 03 00 48 00 1.9...d.|o.c...H.
3410:69 00 6D 00 1B 00 43 00 3A 00 5C 00 57 00 49 00 i.m...C.|:|. |W.I.
3420:4E 00 44 00 4F 00 57 00 53 00 5C 00 44 00 65 00 N.D.O.W.|S|. |D.e.
3430:73 00 6B 00 74 00 6F 00 70 00 5C 00 6C 00 69 00 s.k.t.o.|p|. |l.i.
3440:73 00 74 00 2E 00 64 00 6F 00 63 00 FF 40 01 80 s.t...d.|o.c..@..
```

- ▶ <https://groups.google.com/forum/#!topic/alt.comp.virus/9dnXxxvtUA>

MELISSA VIRUS (1999)

Also I found this GUID in the file:

```
4BC0:5F 50 49 44 5F 47 55 49 44 00 02 00 00 00 E4 04 _PID_GUI|D.....
4BD0:00 00 41 00 00 00 4E 00 00 00 7B 00 35 00 37 00 ..A...N.|..{.5.7.
4BE0:32 00 38 00 35 00 38 00 45 00 41 00 2D 00 33 00 2.8.5.8.|E.A.-.3.
4BF0:36 00 44 00 44 00 2D 00 31 00 31 00 44 00 32 00 6.D.D.-.|1.1.D.2.
4C00:2D 00 38 00 38 00 35 00 46 00 2D 00 30 00 30 00 -.8.8.5.|F.-.0.0.
4C10:34 00 30 00 33 00 33 00 45 00 30 00 30 00 37 00 4.0.3.3.|E.0.0.7.
4C20:38 00 45 00 7D 00 00 00 00 00 00 00 00 00 00 00 8.E.}|...|.....
```

```
8230:47 00 7B 00 33 00 44 00 34 00 35 00 39 00 39 00 G.{.3.D.|4.5.9.9.
8240:36 00 32 00 2D 00 45 00 31 00 42 00 34 00 2D 00 6.2.-.E.|1.B.4.-.
8250:31 00 31 00 44 00 32 00 2D 00 39 00 45 00 42 00 1.1.D.2.|-.9.E.B.
8260:41 00 2D 00 30 00 30 00 34 00 30 00 33 00 33 00 A.-.0.0.|4.0.3.3.
8270:45 00 30 00 30 00 37 00 38 00 45 00 7D 00 23 00 E.0.0.7.|8.E.}|.#.
```

- ▶ <https://groups.google.com/forum/#!topic/alt.comp.virus/9dnXxxxvtUA>

DATA HIDING: CHANGING OR MANIPULATING A FILE TO CONCEAL INFORMATION

DEALING WITH DELETED FILES

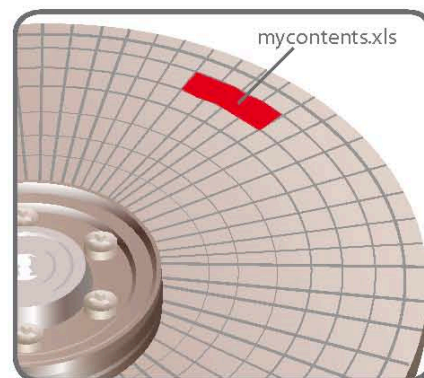
File Carving

- ▶ Reassemble fragments of files scattered across the file system

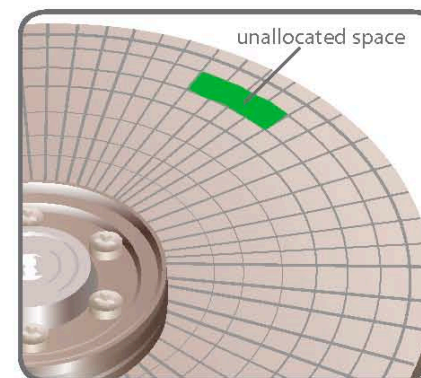
How are Deleted Files and Data Recovered?

Computers Don't Immediately Remove Data that is Deleted

Original Data

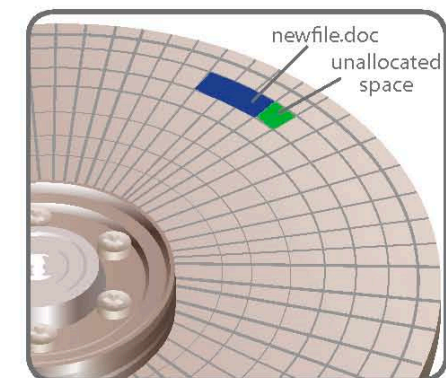


Deleted Data



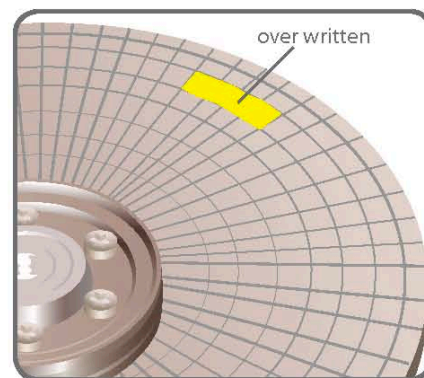
The original data is still present, but marked as unallocated space.

Partially Overwritten Data



Over time, some or all of the data can be over written. The remaining data can still be "carved" and reviewed.

Data Wiped Clean or Shredded



The data can be wiped clean or shredded using privacy software.

What is unallocated space?

Unallocated Space is available disk space that is not allocated to any volume. The type of volume that you can create on unallocated space depends on the disk type. On basic disks, you can use unallocated space to create primary or extended partitions. On dynamic disks, you can use unallocated space to create dynamic volumes.

PINPOINT
LABORATORIES

©2008 Pivotal Guidance

www.pinpointlabs.com

DATA HIDING: CHANGING OR MANIPULATING A FILE TO CONCEAL INFORMATION

- ▶ Changing file names and file extensions
- ▶ Setting file attributes to hidden
- ▶ Hiding entire partitions
- ▶ Bit-shifting
- ▶ Using encryption/password protection

HIDING FILES BY USING THE OS

- One of the first techniques to hide data:
 - Changing file extensions
- Advanced digital forensics tools check file headers
 - Compare the file extension to verify that it's correct
 - If there's a discrepancy, the tool flags the file as a possible altered file
- Another hiding technique
 - Selecting the Hidden attribute in a file's Properties dialog box

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HIDING PARTITIONS

- By using the Windows `diskpart remove letter` command
 - You can unassign the partition's letter, which hides it from view in File Explorer
- To unhide, use the `diskpart assign letter` command
- Other disk management tools:
 - Partition Magic, Partition Master, and Linux Grand Unified Bootloader (GRUB)

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TO DETECT WHETHER A PARTITION HAS BEEN HIDDEN

- Account for all disk space when examining an evidence drive
- Analyze any disk areas containing space you can't account for

Case View Tools Window Help



Add Data Source



Images/Videos



Communications



Timeline



Close Case



Generate Report



Listing

/img_PhysicalDrive0/vol_vol8

Table

Thumbnail

Page:

Pages:



Go to Page:

Name	S	C	Location	Modified Time
\$OrphanFiles			/img_PhysicalDrive0/vol_vol8//\$OrphanFiles	0000-00-00 00:00:0
initrd.img				2018-01-18 08:43:5
initrd.img.old				2017-12-08 11:58:2
vmlinuz				2018-01-18 08:43:5
vmlinuz.20750				2017-12-08 11:58:2
vmlinuz.old				2017-12-08 11:58:2
\$Unalloc			/img_PhysicalDrive0/vol_vol8//\$Unalloc	0000-00-00 00:00:0
[current folder]				2019-07-29 15:20:5
[parent folder]				2019-07-29 15:20:5
bin				2019-07-31 06:52:5
boot				2019-07-31 06:57:3
cdrom				2017-04-12 12:23:5
dev				2017-02-15 15:36:3



Hex

Text

Application

Message

File Metadata

Results

Annotations

Other Occurrences

MARKING BAD CLUSTERS

- A data-hiding technique used in FAT file systems is placing sensitive or incriminating data in free or slack space on disk partition clusters
 - Involves using old utilities such as Norton DiskEdit
- Can mark good clusters as bad clusters in the FAT table so the OS considers them unusable
 - Only way they can be accessed from the OS is by changing them to good clusters with a disk editor
- DiskEdit runs only in MS-DOS and can access only FAT-formatted disk media

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BIT-SHIFTING

- Some users use a low-level encryption program that changes the order of binary data
 - Makes altered data unreadable To secure a file, users run an assembler program (also called a “macro”) to scramble bits
 - Run another program to restore the scrambled bits to their original order
- Bit shifting changes data from readable code to data that looks like binary executable code

UNDERSTANDING STEGANALYSIS METHODS

- Steganography - comes from the Greek word for “hidden writing”
 - Hiding messages in such a way that only the intended recipient knows the message is there
- Steganalysis - term for detecting and analyzing steganography files
- Digital watermarking - developed as a way to protect file ownership
 - Usually not visible when used for steganography

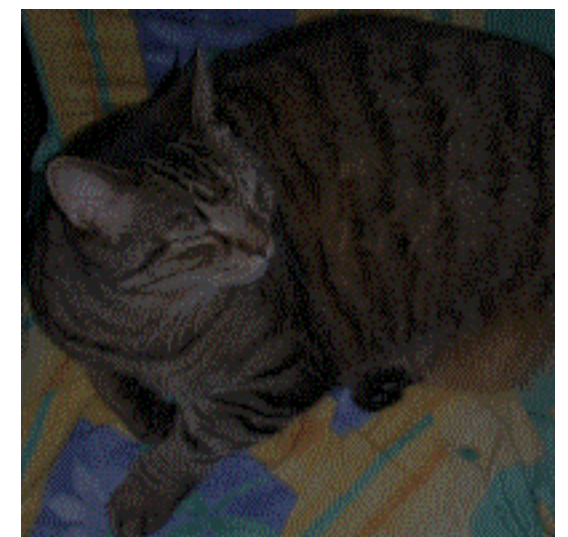
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UNDERSTANDING STEGANALYSIS METHODS

- A way to hide data is to use steganography tools
 - Many are freeware or shareware
 - Insert information into a variety of files
- If you encrypt a plaintext file with PGP and insert the encrypted text into a steganography file
 - Cracking the encrypted message is extremely difficult



Removing all but the two least significant bits of each color component and a subsequent normalization.
<https://en.wikipedia.org/wiki/Steganography>



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EXAMINING ENCRYPTED FILES

- To decode an encrypted file
 - Users supply a password or passphrase
- Many encryption programs use a technology called “key escrow”
 - Designed to recover encrypted data if users forget their passphrases or if the user key is corrupted after a system failure
- Key sizes of 128 bits to 4096 bits make breaking them nearly impossible with current technology

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RECOVERING PASSWORDS

- Password-cracking tools are available for handling password-protected data or systems
 - Some are integrated into digital forensics tools
- Stand-alone tools:
 - Last Bit
 - AccessData PRTK
 - ophcrack
 - John the Ripper
 - Passware

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RECOVERING PASSWORDS

- Brute-force attacks
 - Use every possible letter, number, and character found on a keyboard
 - This method can require a lot of time and processing power
- Dictionary attack
 - Uses common words found in the dictionary and tries them as passwords
 - Most use a variety of languages

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RECOVERING PASSWORDS

- With many programs, you can build profiles of a suspect to help determine his or her password
- Many password-protected OSs and application store passwords in the form of hash values
- A brute-force attack requires converting a dictionary password from plaintext to a hash value
 - Requires additional CPU cycle time

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RECOVERING PASSWORDS

- Rainbow table
 - A file containing the hash values for every possible password that can be generated from a computer's keyboard
 - No conversion necessary, so it is faster than a brute-force or dictionary attack
- Salting passwords
 - Alters hash values and makes cracking passwords more difficult



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ACTIVITY

Graphical Timeline of the Data Leakage Scenario

